FIRE: THE FIRST-YEAR INNOVATION & RESEARCH EXPERIENCE

RAPID DIAGNOSTICS

# Detecting E. Coli in Fast Foods

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## Introduction

**Background**

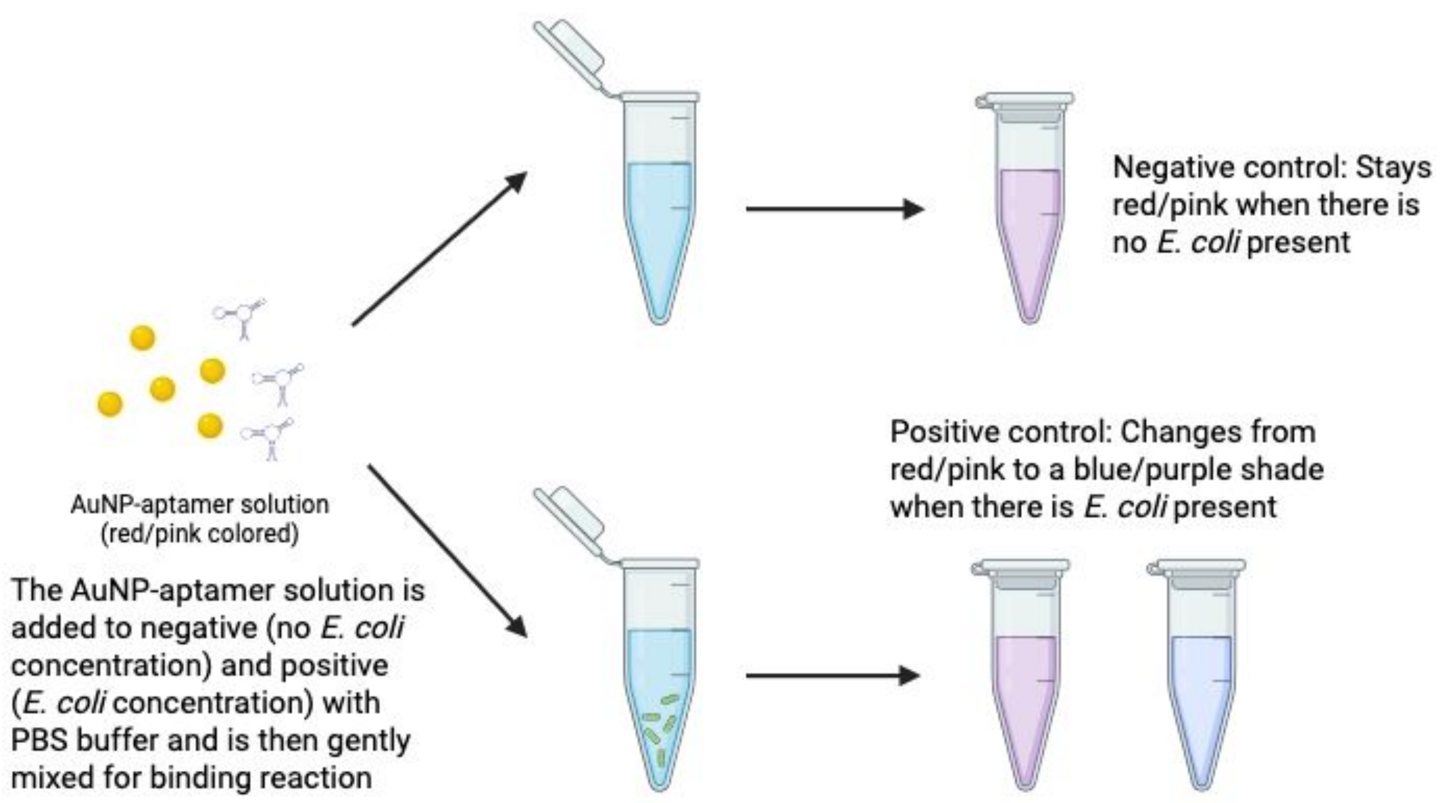
- *Escherichia coli* (*E. coli*) is a kind of bacteria found in the environment and human and animal intestines.
- Bacteria like *E. coli* are known to cause major health issues like urinary tract infections and kidney failures, sometimes leading to hospitalization and in severe cases, death.
- *E. coli* outbreaks are fairly common, especially in fast food chains.

**Methods and Problems**

- The thiol DNA aptamer is bound to the gold nanoparticles (AuNPs) which recognize and bind to the *E. coli* cells, causing a signal change through colorimetric detection methods.
- Current biosensors for the general public that rapidly detect *E. coli* are costly and inefficient, and take a long time to generate results.

**Approach: Developing a fast, inexpensive, and efficient colorimetric device to detect *E. coli* in fast foods by binding gold nanoparticles to an *E. coli* aptamer to trigger a color change.**

## Methods




**Colorimetric Device Setup**

- The colorimetric device is initially set up with the $1*10^6$ CFU/mL concentration of *E. coli* diluted with the phosphate-buffered solution (PBS).
- The gold nanoparticles (AuNPs) and aptamer solution is also prepared for the binding reaction.

**Reaction Mechanism**

- When the AuNP-aptamer complex is added to the *E. coli* and buffer solution, the aptamer recognizes the *E. coli* particles.
- A network is created, allowing the gold nanoparticles to bind to the *E. coli* through the aptamer as they are in close proximity.
- The binding process should change the color of the solution from red/pink to a blue/purple shade when *E. coli* is present.
- We also utilized a salt, NaCl, to help allow aggregation of AuNPs[1].

## Results/Discussion

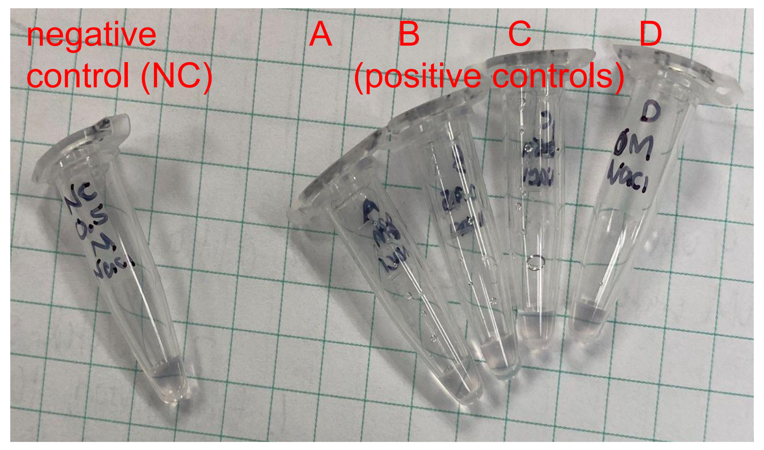



**Figure 1.** Varying concentrations of NaCl (NC: 0.5M, A: 1.0M, B: 0.5M, C: 0.25M, and D: 0M). A-D have AuNP-aptamer solution and *E. coli* and the NC has AuNP-aptamer solution but no *E.coli*.

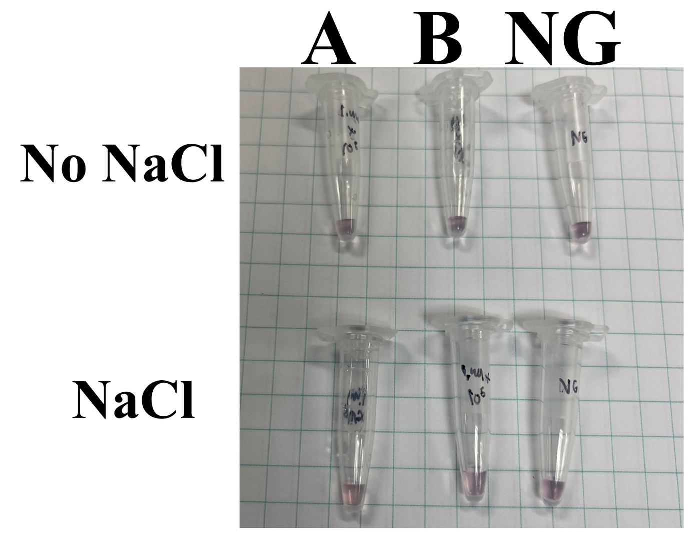



**Figure 2.** Test of 10 µL AuNP-aptamer solution with either 5 µL of 1M NaCl or 5 µL of HEPES buffer in A, B, and C. 10 µL of *E.coli* was added with concentrations of $9.44 \times 10^9$ CFU/mL and $9.44 \times 10^6$ CFU/mL in A and B, respectively. NG (negative control) has 10 µL HEPES buffer instead of *E.coli*.

**Synthesis of AuNP-Aptamer Conjugates**

- Initially, the DNA-AuNP conjugates were washed with deionized (DI) water, which may have caused the first few AuNP-aptamer solutions to aggregate[2], as seen in **Figure 1.**
- We washed the final DNA-AuNP conjugates with HEPES buffer, replacing the water, which may have helped in reduced aggregation of the gold nanoparticles in the AuNP-aptamer solution, due to the stronger buffering capabilities reducing background noise.

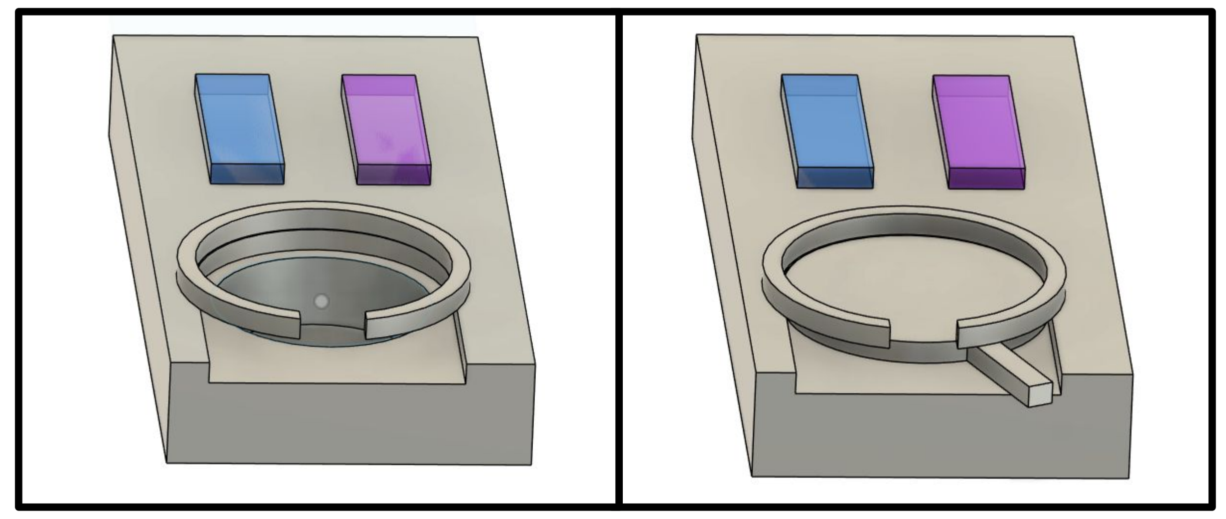
**Optimization of Binding Conditions**

- We used varying amounts of *E. coli*, phosphate-buffered saline, and salt concentrations for binding.
  - We found that increasing the concentration of *E. coli* was the most effective in producing a visible color change and difference between the positive and negative controls.
  - However, our samples were purple at lower concentrations and in the negative control and they were red for higher concentrations, which is backwards from what is expected, as seen in **Figure 2.** This may be attributed to extremely high bacterial concentrations stabilizing the AuNP-aptamer solution, turning it back to red.

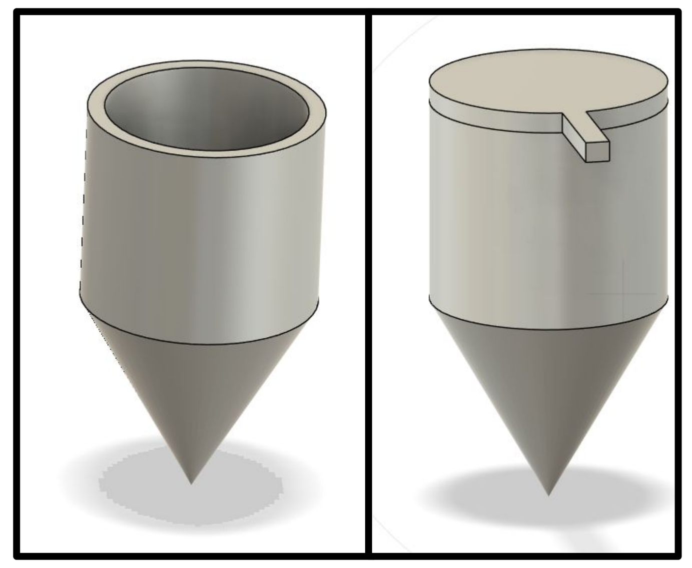
## Conclusion/Future Directions

Our project demonstrates the feasibility of a rapid, low-cost colorimetric biosensor for *E. coli* detection using AuNP aptamer conjugates. Optimizing washing buffers and salt concentrations significantly improved sensor performance by reducing background aggregation and enhancing signal clarity. With further calibration, this platform could be adapted into a microfluidic-based device capable of on-site testing in food industry settings. In the long term, the goal is to produce a commercially viable, point-of-care diagnostic device suitable for use in fast-food kitchens and field inspections. A possible future adjustment to make this more suitable for that would include lowering the limit of detection (LOD) below $10^7$ CFU/mL by refining aptamer density and reaction time.

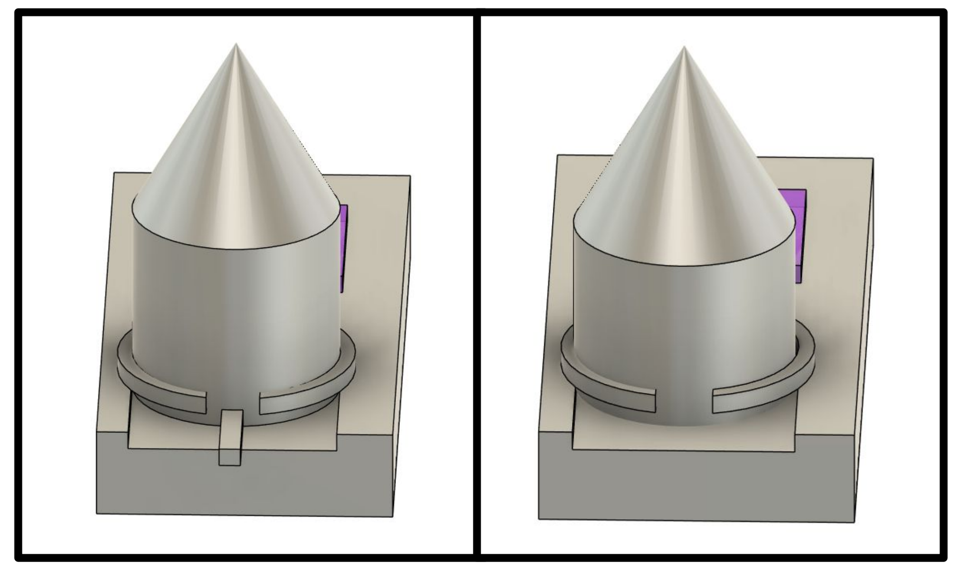
## Device Concept



**STEP 1:** Remove the cover from device by pulling the tag horizontally.



**STEP 2:** Put the *E.coli* sample in the tube and close it using the lid.



**STEP 3:** Holding the lid tightly with your thumb and the bottom with your finger(s), insert it into the device from above, fitting the tag through the gap, then pull the tag out slowly.

## Acknowledgements

- Methods figure created in biorender.com
- We thank the FIRE Rapid Diagnostics stream, the Office of Undergraduate Research at the University of Maryland, and Dr. Shannon Hilton for their mentorship, guidance, and laboratory support.

## References

1. Xie, Y. et al, *Sensors and Actuators B: Chemical*, 2021, 339.
2. Giorgi-Coll, S. et al, *Mikrochimica acta*, 2019, 187(1), 13.
3. Ding, Y. et al, *Chinese Journal of Chemistry,* 2024, *42*(19), 2391-2400.